SINGLE-CENTRED DYON INDICES OF THE $\mathbb{Z}_2$ CHL MODEL ARE POSITIVE AT $Q^2 \leq 2$

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Abstract. In the $\mathbb{Z}_2$ CHL model of $\mathcal{N} = 4$ string theory we prove Sen's conjecture that the index of a single-centred quarter-BPS black hole is positive, for every charge with $Q^2 = 1$ or $Q^2 = 2$ and positive discriminant and for every charge in its duality orbit. The proof writes the relevant mock Jacobi forms through theta functions, Eisenstein series and Hurwitz class numbers and reads off the sign of each coefficient.

Keywords: CHL model; dyon degeneracy; single-centred black hole; mock Jacobi form; Hurwitz class number.

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1. Introduction

A quarter-BPS dyon of $\mathcal{N} = 4$ string theory in four dimensions carries an electric charge Q and a magnetic charge P, and the protected index that counts such states is the helicity trace B_6. When the charges admit a single-centred black hole whose only hair are the zero modes of the broken supersymmetries, the twelve fermion zero modes fix the sign of the trace, and the horizon degrees of freedom contribute $-B_6 > 0$. Sen turned this observation into a conjecture about Fourier coefficients of Siegel modular forms [16]: the index $d^*(Q, P)$ of the single-centred black hole, extracted from the inverse of the dyon partition function with the contour of the attractor region, is positive.[1]

For type II string theory on $K3 \times T^2$, where the partition function is Φ_{10} [10], the conjecture has been settled in two steps. Bringmann and Murthy proved it for the two lowest magnetic charges $P^2 = 2, 4$ by writing the finite parts of the first two Fourier–Jacobi coefficients of $1/\Phi_{10}$ through class numbers [3], using the decomposition into polar and finite parts of Dabholkar, Murthy and Zagier [8]. Rosselló then proved positivity for all charges [13]. The CHL models [5] are the $\mathbb{Z}_N$ orbifolds of type IIB on $K3 \times T^2$ in which the $\mathbb{Z}_N$ acts on $K3$ by a symplectic automorphism and on T^2 by a shift; their partition functions $\widetilde{\Phi}_k$ are Siegel modular forms of weight k for subgroups of $Sp(2, \mathbb{Z})$ [12, 7, 9].[2] Bhand, Sen and Singh showed that the Fourier–Jacobi coefficients of $1/\widetilde{\Phi}_k$ are meromorphic Jacobi forms, split them into an Appell–Lerch polar part and a mock Jacobi finite part, and identified the single-centred index with the Fourier coefficients of the finite part on an explicit set of charges [4, Theorem 5.4]. The positivity conjecture for the CHL models remains open; Cardoso, Nampuri and Rosselló gave an exact formula for the immortal dyon index in these models [6], and Singh constructed a generating function for the single-centred index [17], but neither proves the sign.

[1] Sen checked the sign on the charges of his Tables 1 to 5, for the heterotic string on T^6 and for the CHL models of orders $2, 3, 5, 7$, and stated the conjecture for all charges in the cones (3.7)–(3.12) of [16].

[2] For $N = 2$ the model is the heterotic string on T^6 obtained from the $E_8 \times E_8$ string by exchanging the two E_8 factors together with a half shift on a circle [5]; for the ten-dimensional $E_8 \times E_8$ theory and its relation to M-theory see [14]. Background on the involutions of $K3$ surfaces and on Calabi–Yau threefolds fibred by $K3$ surfaces, which underlie the dual type II description, is in [1, 2].

This paper proves the conjecture for the $\mathbb{Z}_2$ CHL model on the first two rows of electric charge. The model has $k = 6$, duality group $\Gamma_1(2)$, and charges normalised so that Q^2, $P^2/2$ and $Q \cdot P$ are integers.

Theorem A. *In the $\mathbb{Z}_2$ CHL model the single-centred index $d^*(Q,P)$ is positive for every charge with $Q^2 \in \{1,2\}$, $P^2 > 0$ and $Q^2P^2 - (Q\cdot P)^2 > 0$, and for every charge in the $\Gamma_1(2)$ orbit of such a charge.*

The case $Q^2 = 1$ is Theorem 3.3 and the case $Q^2 = 2$ is Theorem 4.8. In both cases the proof has the shape of [3]. A product formula identity puts the Fourier–Jacobi coefficient φ_m of $1/\widetilde{\Phi}_6$ into closed form; a theorem of Dabholkar, Murthy and Zagier, or for $m = 1$ its special case due to Bringmann and Murthy, separates the polar part and expresses the mock part of the finite part through Hurwitz class numbers; and an analysis of the theta components of the remaining holomorphic Jacobi form gives the sign of every coefficient. The new ingredient at $Q^2 = 2$ is a factorisation (Proposition 4.5) of that holomorphic part as

$$J_2 = \frac{16\,\theta_4(\sigma,v)^2}{\theta_4(\sigma)^6}\Big[\theta_2(\sigma)^2\big(\theta_3(\sigma)^4 + 3\theta_2(\sigma)^4\big)\theta_2(\sigma,v)^2 - \theta_3(\sigma)^2\big(3\theta_3(\sigma)^4 + \theta_2(\sigma)^4\big)\theta_3(\sigma,v)^2\Big],$$

from which the signs of its theta components are manifest (Lemma 4.6).

The index does not decide its own sign by a general principle; what makes the two rows tractable is that the weak Jacobi forms of index one and two are few, so the finite part is a short combination of explicit functions. The third row $Q^2 = 3$ has the same structure, and we describe at the end what is known there (Remark 5.1). The full conjecture for the $\mathbb{Z}_2$ CHL model, for all Q^2, and the conjecture for the models of order $3, 5, 7$ remain open.

Section 2 recalls the partition function, the decomposition of Bhand, Sen and Singh and the auxiliary functions. Section 3 treats $Q^2 = 1$ and Section 4 treats $Q^2 = 2$. Section 5 compares the first values with the product formula and with Sen's table.

2. The partition function and the single-centred index

2.1. **Notation.** Write $\mathbf{e}(t) = e^{2\pi i t}$. The variables of the Siegel upper half-space are (ρ,σ,v); put $x = \mathbf{e}(\sigma/2)$, $p = x^2 = \mathbf{e}(\sigma)$ and $y = \mathbf{e}(v)$. For a function of σ alone we use the expansion in x. The Jacobi theta functions are

$$\theta_1(\tau,z) = -i\sum_{r\in\mathbb{Z}+\frac12}(-1)^{r-\frac12}q^{r^2/2}\zeta^r, \qquad \theta_2(\tau,z) = \sum_{r\in\mathbb{Z}+\frac12} q^{r^2/2}\zeta^r,$$

$$\theta_3(\tau,z) = \sum_{n\in\mathbb{Z}} q^{n^2/2}\zeta^n, \qquad \theta_4(\tau,z) = \sum_{n\in\mathbb{Z}}(-1)^n q^{n^2/2}\zeta^n,$$

with $q = \mathbf{e}(\tau)$, $\zeta = \mathbf{e}(z)$, and $\theta_k(\tau) = \theta_k(\tau,0)$. Let $E(\tau) = 1 - 24\sum\sigma_1(n)q^n$, $E_4 = 1 + 240\sum\sigma_3(n)q^n$, $E_6 = 1 - 504\sum\sigma_5(n)q^n$, and let

$$\mathfrak{E}(\tau) = 2E(2\tau) - E(\tau) = 1 + 24\sum_{n\ge1}\sigma_1^{\mathrm{odd}}(n)q^n, \qquad \sigma_1^{\mathrm{odd}}(n) = \sum_{d\mid n,\ d\ \mathrm{odd}} d,$$

be the Eisenstein series of weight two for $\Gamma_0(2)$. The weak Jacobi forms of weight -2 and 0 and index one are

$$\phi_{-2,1}(\tau,z) = -\frac{\theta_1(\tau,z)^2}{\eta(\tau)^6} = \zeta - 2 + \zeta^{-1} + O(q), \qquad \phi_{0,1}(\tau,z) = 4\sum_{k=2}^{4}\frac{\theta_k(\tau,z)^2}{\theta_k(\tau)^2} = \zeta + 10 + \zeta^{-1} + O(q)$$

[11, §9]; we write $A=\phi_{-2,1}(\sigma,v)$ and $B=\phi_{0,1}(\sigma,v)$. The Hurwitz class number $H(D)$ is extended by $H(0)=-\frac{1}{12}$ and $H(D)=0$ unless D is a nonnegative integer with $D\equiv 0,3 \pmod 4$. Put $g(\tau)=\eta(\tau)^8\eta(2\tau)^8$ and

$$\frac{1}{g(\sigma/2)}=\sum_{j\ge 0}\beta(j)\,x^{j-1},\qquad \sum_{j\ge 0}\beta(j)x^j=\prod_{r\ge 1}(1-x^r)^{-e_r},\quad e_r=\begin{cases}8, & r \text{ odd},\\ 16, & r \text{ even},\end{cases} \tag{1}$$

so that every $\beta(j)$ is a positive integer.

2.2. **The product formula.** The twisted elliptic genera of $K3$ for the $\mathbb{Z}_2$ symplectic involution are [7]

$$\begin{aligned} F^{(0,0)}&=\phi_{0,1},\qquad F^{(0,1)}=\tfrac{1}{3}\phi_{0,1}+\tfrac{2}{3}\phi_{-2,1}\,\mathfrak{E}(\tau),\\ F^{(1,0)}&=\tfrac{1}{3}\phi_{0,1}-\tfrac{1}{3}\phi_{-2,1}\,\mathfrak{E}(\tfrac{\tau}{2}),\qquad F^{(1,1)}=\tfrac{1}{3}\phi_{0,1}-\tfrac{1}{3}\phi_{-2,1}\,\mathfrak{E}(\tfrac{\tau+1}{2}),\end{aligned} \tag{2}$$

each a weak Jacobi form of weight 0 and index one for $\Gamma(2)$, and $SL_2(\mathbb{Z})$ permutes them.[3] Write $F^{(r,s)}(\tau,z)=\sum_{b\in\{0,1\}}\sum_{j\in 2\mathbb{Z}+b}\sum_{n\in\mathbb{Z}/2}c_b^{(r,s)}(4n-j^2)q^n\zeta^j$. The dyon partition function of the model is [12, 7, 9]

(3)

$$\widetilde{\Phi}_6(\rho,\sigma,v)=\mathbf{e}\big(\rho+\tfrac{\sigma}{2}+v\big)\prod_{b=0}^{1}\prod_{r=0}^{1}\prod_{\substack{k'\in\mathbb{Z}+r/2,\ l\in\mathbb{Z},\ j\in 2\mathbb{Z}+b\\ k',l\ge 0;\ j<0 \text{ if } k'=l=0}}\big(1-\mathbf{e}(k'\sigma+l\rho+jv)\big)^{\sum_{s=0}^{1}(-1)^{sl}c_b^{(r,s)}(4k'l-j^2)},$$

and the index of the charges (Q,P) is $(-1)^{Q\cdot P+1}$ times the coefficient of $\mathbf{e}(m\rho)x^n y^\ell$ with

$$m=Q^2,\qquad n=P^2/2,\qquad \ell=Q\cdot P, \tag{4}$$

in the expansion of $1/\widetilde{\Phi}_6$ on the appropriate contour [10, 12, 15].[4]

Expand $1/\widetilde{\Phi}_6=\sum_{m\ge -1}\varphi_m(\sigma,v)\,\mathbf{e}(m\rho)$. The factors with $l=0$ in (3) have the exponents $\sum_s c_b^{(r,s)}(-j^2)$, equal to $12,2,8,0$ for $(r,|j|)=(0,0),(0,1),(1,0),(1,1)$ and to 0 for $|j|\ge 2$; together with the prefactor they give

$$\varphi_{-1}=\frac{1}{g(\sigma/2)\,A}, \tag{5}$$

since $\phi_{-2,1}=-\theta_1^2/\eta^6$. Taking the logarithm of the factors with $l\ge 1$ and collecting the powers of $\mathbf{e}(\rho)$,

$$\begin{aligned}\frac{1}{\widetilde{\Phi}_6}&=\varphi_{-1}\,\mathbf{e}(-\rho)\exp\Big(\sum_{l\ge 1}L_l(\sigma,v)\,\mathbf{e}(l\rho)\Big),\\ L_l&=\sum_{td=l}\frac{1}{2dt}\sum_{r,s\in\{0,1\}}(-1)^{sd}\sum_{u=0}^{2d-1}(-1)^{ru}F^{(r,s)}\Big(\frac{t\sigma+u}{d},\,tv\Big).\end{aligned} \tag{6}$$

The sum over u keeps exactly the integral powers of x, so each L_l is a series in x whose coefficients are Laurent polynomials in y; in particular L_l is entire in v. Hence

$$g(\tfrac{\sigma}{2})\,A\,\varphi_1=L_2+\tfrac{1}{2}L_1^2,\qquad g(\tfrac{\sigma}{2})\,A\,\varphi_2=L_3+L_1L_2+\tfrac{1}{6}L_1^3. \tag{7}$$

[3] In the normalisation of [7], $F^{(r,s)}=\frac{1}{N}\operatorname{Tr}_{RR,g^r}\big(g^s(-1)^{F_L+F_R}q^{L_0}\bar{q}^{\bar{L}_0}y^J\big)$, so that $F^{(0,0)}$ is half the elliptic genus $2\phi_{0,1}$ of $K3$; the constant terms then give the exponents $12,2,8,0$ used below.

[4] The exponent of $\mathbf{e}(\rho)$ is $NQ^2/2=Q^2$, and the power of $x=\mathbf{e}(\sigma/2)$ is $P^2/2$, as in [4, (5.15)].

Lemma 2.1. *The functions L_1, L_2, L_3 are invariant under $\Gamma(2)$, $\Gamma(4)$, $\Gamma(6)$ respectively, acting on σ, and L_l has the elliptic property of index l in v. Every polynomial in L_1, L_2, L_3 that is homogeneous of total index I is a Jacobi form of weight 0 and index I for $\Gamma(12)$, even in v and weak at every cusp.*

Proof. Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$ and $M = \left(\begin{smallmatrix} t & u \\ 0 & d \end{smallmatrix}\right)$. Then

$$M\gamma M^{-1} = \begin{pmatrix} a + uc/t & \big(tb + u(e-a) - u^2 c/t\big)/d \\ dc/t & e - uc/t \end{pmatrix}.$$

For $(t,d) = (1,1)$ and $\gamma \in \Gamma(2)$, for $(t,d) = (1,2), (2,1)$ and $\gamma \in \Gamma(4)$, and for $(t,d) = (1,3), (3,1)$ and $\gamma \in \Gamma(6)$, this matrix lies in $\Gamma(2)$, under which every $F^{(r,s)}$ is invariant. This gives the invariance, and $\Gamma(12)$ is contained in all three groups. Under $v \mapsto v + \lambda\sigma + \mu$ the variable tv of the term (t,d,u) moves by $\lambda d\,\tau' + (t\mu - \lambda u)$ with $\tau' = (t\sigma + u)/d$, and the elliptic property of index one of $F^{(r,s)}$ produces the factor $\mathbf{e}(-\lambda^2 d\tau' - 2\lambda d\, tv) = \mathbf{e}(-\lambda^2 l\sigma - 2\lambda l v)$, which is the elliptic property of index l. Evenness in v is inherited from the genera. For $\gamma \in SL_2(\mathbb{Z})$ one has $M\gamma = \gamma' M'$ with $\gamma' \in SL_2(\mathbb{Z})$ and M' upper triangular of the same determinant, and $SL_2(\mathbb{Z})$ permutes the $F^{(r,s)}$, none of which contains a negative power of q; so every term is weak at every cusp. □

2.3. **Polar part, finite part and the single-centred index.** By [4, Theorem 5.2], φ_m is a meromorphic Jacobi form of weight -6 and index m for $\Gamma_1(2)$ with double poles at $v \in \mathbb{Z}\sigma + \mathbb{Z}$. Its polar part is [4, (5.45)]

$$\varphi_m^P = \frac{\hat f(m)}{g(\sigma/2)}\,\mathcal{A}_{2,m}, \qquad \mathcal{A}_{2,m}(\sigma, v) = \sum_{s\in\mathbb{Z}} \frac{p^{ms^2+s} y^{2ms+1}}{(1-p^s y)^2}, \tag{8}$$

where $\hat f(m)$ is the coefficient of q^m in $1/\big(\eta(\tau)^8 \eta(2\tau)^8\big)$, so that $\hat f(-1), \hat f(0), \hat f(1), \hat f(2), \hat f(3) = 1, 8, 52, 256, 1122$. The finite part $\varphi_m^F = \varphi_m - \varphi_m^P = \sum c_m^F(n,\ell) x^n y^\ell$ is a mock Jacobi form.

Let $\mathcal{A}$ be the set of charges of [4, (5.46)–(5.47)]: with $N = 2$ and the notation (4), the charges with $m, n > 0$ and

$$D := Q^2 P^2 - (Q \cdot P)^2 = 2mn - \ell^2 > 0, \tag{9}$$

such that

$$\min_{m_1 \in 2\mathbb{Z},\ s\in\mathbb{Z},\ m_1 \mid s(s+1)} \Big[\big(m_1 \ell + 2ms + m\big)^2 + m_1^2 D \Big] \ \geq\ m^2. \tag{10}$$

By [4, Theorem 5.4(a)], for every charge in $\mathcal{A}$,

$$d^*(Q,P) = (-1)^{\ell+1} c_m^F(n,\ell). \tag{11}$$

The single-centred index is invariant under the action [4, (5.17)–(5.19)] of $\Gamma_1(2)$ on the charges.

Lemma 2.2. *Every charge with $Q^2 \in \{1,2\}$, $P^2 > 0$ and $D > 0$ lies in $\mathcal{A}$.*

Proof. If $m_1 = 0$, the condition $m_1 \mid s(s+1)$ reads $s(s+1) = 0$, and the bracket is $m^2(2s+1)^2 = m^2$. If $m_1 \neq 0$, then $m_1^2 \geq 4 \geq m^2$ and $D \geq 1$, so the bracket is at least m^2. □

So for $Q^2 \in \{1,2\}$ the theorem is the statement that $(-1)^{\ell+1} c_m^F(n,\ell) > 0$ whenever $D > 0$.

2.4. **A positivity lemma.** The last step in both cases divides a series of known sign pattern by g.

Lemma 2.3. *Let $f = -1+\sum_{n\ge1} a_n x^n$ with $a_n > K$ for every n, and let $0 < e_r \le K$ for every $r \ge 1$. Then $f\prod_{r\ge1}(1-x^r)^{-e_r}$ has positive coefficients at every x^n with $n \ge 1$.*

Proof. Let $f_0 = f$ and $f_s = f_{s-1}(1-x^s)^{-e_s} = -1 + \sum_n b_n^{(s)} x^n$. We show by induction on s that $b_n^{(s)} > 0$ for $n \ge 1$ and $b_n^{(s)} > K$ for $n > s$; since f_s and the full product agree through x^s, this proves the lemma. The case $s = 0$ is the hypothesis. Write $e = e_s$ and $b = b^{(s-1)}$. If $s \nmid n$, then $b_n^{(s)} = \sum_k \binom{e+k-1}{k} b_{n-sk} \ge b_n$, all terms being positive. If $n = sm$, then

$$b_{sm}^{(s)} = \sum_{k=0}^{m-1} \binom{e+k-1}{k} b_{s(m-k)} - \binom{e+m-1}{m} > \binom{e+m-1}{m-1}\Big(K - \frac{e}{m}\Big),$$

since every $b_{s(m-k)}$ exceeds K and $\sum_{k=0}^{m-1}\binom{e+k-1}{k} = \binom{e+m-1}{m-1}$. The right side is $K - e \ge 0$ for $m = 1$, and at least $(e+1)(K-\frac{e}{2}) \ge (e+1)\frac{K}{2} \ge K$ for $m \ge 2$. □

We also use the elementary bound $H(D) \le D$ for $D \ge 3$: a reduced form (a,b,c) of discriminant $-D$ has $|b| \le a \le \sqrt{D/3}$ and is determined by (a,b), so $H(D) \le \sqrt{D/3}\,(2\sqrt{D/3}+1) \le D$.

3. The first row: $Q^2 = 1$

Let $\mathcal{H}(\sigma,v) = \sum_{n,\ell} H(4n-\ell^2)p^n y^\ell$ be the generating function of the Hurwitz class numbers, $\mathcal{B} = \mathcal{A}_{2,1}$, $r_3(D)$ the number of representations of D as a sum of three squares,

$$M(\tau) = \frac{E_4(\tau) - E_4(2\tau)}{240} = \sum_{n\ge1} \sigma_3^{\mathrm{odd}}(n) q^n, \qquad \sigma_3^{\mathrm{odd}}(n) = \sum_{d\mid n,\ n/d \text{ odd}} d^3,$$

$$\Theta(\sigma,v) = \sum_{e,\ell} r_3(2e-\ell^2)\, x^e y^\ell.$$

Then $\Theta = \frac12\big[\theta_3(\frac{\sigma}{2},v)\theta_3(\frac{\sigma}{2})^3 + \theta_4(\frac{\sigma}{2},v)\theta_4(\frac{\sigma}{2})^3\big]$ is the theta series of the lattice D_4, with the Jacobi variable on one coordinate vector, at $\tau = \sigma/2$.[5] Write $\phi_{-2,1}(\tau,z) = \sum \alpha(4a-r^2)q^a\zeta^r$, so that $\alpha(-1) = 1$ and $\alpha(0) = -2$.

Proposition 3.1. *In the $\mathbb{Z}_2$ CHL model*

$$g(\tfrac{\sigma}{2})\,\varphi_1 = 52\,\mathcal{B} + 48\,M(\tfrac{\sigma}{2})\,A - \tfrac{20}{3}\,\Theta - 104\,\mathcal{H}. \tag{12}$$

Consequently $c_1^F(n,\ell)$ depends only on $D = 2n-\ell^2$, and

$$\begin{aligned} c_1^F(n,\ell) &= \mathcal{C}(D) := \sum_{j\ge0} \beta(j)\,\mathcal{N}(D+2-2j), \\ \mathcal{N}(D) &= 48\sum_{t\ge1} \sigma_3^{\mathrm{odd}}(t)\,\alpha(D-2t) - \tfrac{20}{3}\,r_3(D) - 104\,H(D). \end{aligned} \tag{13}$$

Proof. Step 1. We claim that

$$L_2 + \tfrac12 L_1^2 = \tfrac{13}{36}\,B^2 - \tfrac{5}{9}\,\mathfrak{E}(\tfrac{\sigma}{2})\,AB + \Big(\tfrac{4}{45}E_4(\tfrac{\sigma}{2}) + \tfrac{19}{180}E_4(\sigma)\Big)A^2. \tag{14}$$

[5] A vector of D_4 is an integral vector of even coordinate sum; the term $\frac12[\cdots]$ selects these, and its norm is $2e$ when the first coordinate is ℓ and the others contribute $2e-\ell^2$.

Every term is a Jacobi form of weight 0 and index two for $\Gamma(4)$, even and holomorphic in v and weak at every cusp (Lemma 2.1; $\mathfrak{E}(\sigma/2)$ and $E_4(\sigma/2)$ are modular for $\Gamma^0(2)\supset\Gamma(4)$). Let G be the difference of the two sides and $G=\chi_0(\sigma)+\chi_2(\sigma)v^2+\chi_4(\sigma)v^4+\cdots$. If $\chi_0=\cdots=\chi_{\nu-2}=0$, the lowest order in v of the transformation law shows that χ_ν is a modular form of weight ν for $\Gamma(4)$ [11, §3], holomorphic at the cusps because G is weak. Since $[PSL_2(\mathbb{Z}):\bar\Gamma(4)]=24$ and the width of the cusp ∞ is four, Sturm's theorem [18] shows that such a form vanishes once its coefficients of $x^0,\dots,x^\nu$ do. The coefficients of G, computed from (6) and (2), vanish through x^{48}, so $\chi_0=\chi_2=\chi_4=0$. For each σ the entire function $G(\sigma,\cdot)$ then vanishes to order at least six at $v=0$, while a nonzero function with the elliptic property of index two has exactly four zeros in a period parallelogram [11, Theorem 1.2]. Hence $G=0$.

Step 2. Bringmann and Murthy [3, Section 3] proved

$$\frac{B^2}{A}=144\,\mathcal{B}+E_4(\sigma)\,A-288\,\mathcal{H}, \tag{15}$$

and the identity

$$\mathfrak{E}(\tfrac{\sigma}{2})\,B=\tfrac15\big(6E_4(\sigma)-E_4(\tfrac{\sigma}{2})\big)A+12\,\Theta \tag{16}$$

holds because both sides are holomorphic Jacobi forms of weight two and index one for $\Gamma^0(2)$ [11, §7]: the argument of Step 1, with $[PSL_2(\mathbb{Z}):\bar\Gamma^0(2)]=3$, width two, Sturm bound $x^{\lfloor\nu/4\rfloor}$ and two zeros in a period parallelogram, reduces it to the coefficients through x^{48}, which agree.

Step 3. Divide (14) by A and insert (15) and (16). The coefficient of $\mathcal{B}$ is $\frac{13}{36}\cdot144=52$, that of $\mathcal{H}$ is $-\frac{13}{36}\cdot288=-104$, that of Θ is $-\frac59\cdot12=-\frac{20}{3}$, and that of A is

$$\Big(\tfrac{13}{36}-\tfrac23+\tfrac{19}{180}\Big)E_4(\sigma)+\Big(\tfrac19+\tfrac{4}{45}\Big)E_4(\tfrac{\sigma}{2})=\tfrac15\big(E_4(\tfrac{\sigma}{2})-E_4(\sigma)\big)=48\,M(\tfrac{\sigma}{2}).$$

With (7) this is (12). The term $52\mathcal{B}/g(\sigma/2)$ is the polar part (8), and the remaining numerator has the coefficient $\mathcal{N}(2e-\ell^2)$ at x^ey^ℓ; multiplying by (1) gives (13). □

Lemma 3.2. (i) $\alpha(D)>0$ *for* $D\equiv3\pmod4$, $\alpha(D)<0$ *for* $D\equiv0\pmod4$ *with* $D\geq0$, *and* $\alpha(D)=0$ *otherwise.*

(ii) $\mathcal{N}(D)=0$ *for* $D<0$, $\mathcal{N}(0)=2$, $\mathcal{N}(D)>0$ *for odd* $D\geq1$, *and* $-\mathcal{N}(2t)\geq96\,\sigma_3^{\rm odd}(t)$ *for* $t\geq1$.

Proof. (i) With $\vartheta_{1,\mu}(\tau,z)=\sum_{r\equiv\mu\,(2)}q^{r^2/4}\zeta^r$ one has $\phi_{-2,1}=\big[\vartheta_{1,0}(\tau,0)\vartheta_{1,1}(\tau,z)-\vartheta_{1,1}(\tau,0)\vartheta_{1,0}(\tau,z)\big]/\eta(\tau)^6$ [3, (4.1)]. So $\sum_{D\equiv3\,(4)}\alpha(D)q^{(D+1)/4}=\vartheta_{1,0}(\tau,0)\prod(1-q^n)^{-6}$ and $\sum_{D\equiv0\,(4)}\alpha(D)q^{D/4}=-q^{-1/4}\vartheta_{1,1}(\tau,0)\prod(1-q^n)^{-6}$, and both right sides have positive coefficients at every exponent ≥0.

(ii) For $D<0$ every argument $D-2t$ is at most -3, and $r_3(D)=H(D)=0$. At $D=0$, $\mathcal{N}(0)=-\frac{20}{3}+\frac{104}{12}=2$. For $D=2t$ every $\alpha(2t-2t')$ is ≤0 by (i), the term $t'=t$ is $\alpha(0)=-2$, and $r_3,H\geq0$; this gives $-\mathcal{N}(2t)\geq96\,\sigma_3^{\rm odd}(t)$. For odd D every $\alpha(D-2t)$ is ≥0, and the term $t=(D+1)/2$ is $\alpha(-1)\sigma_3^{\rm odd}(t)\geq t^3$, so the sum in $\mathcal{N}(D)$ is at least $6(D+1)^3$. Further $r_3(D)\leq(2\sqrt{D}+1)\max_{k\leq D}r_2(k)\leq(2\sqrt{D}+1)\,8\sqrt{D}\leq24D$, from $r_2(k)\leq4d(k)\leq8\sqrt{k}$, and $H(D)\leq D$. Hence $\mathcal{N}(D)\geq6(D+1)^3-264D>0$ for odd $D\geq7$, while $\mathcal{N}(1),\mathcal{N}(3),\mathcal{N}(5)=8,296,1568$. □

Theorem 3.3. *In the* $\mathbb{Z}_2$ *CHL model,* $d^*(Q,P)>0$ *for every charge with* $Q^2=1$, $P^2>0$ *and* $P^2>(Q\cdot P)^2$, *and for every charge in the* $\Gamma_1(2)$ *orbit of such a charge.*

Proof. By Lemma 2.2, (11) and Proposition 3.1, $d^*=(-1)^{\ell+1}\mathcal{C}(D)$ with $D=2n-\ell^2\equiv\ell\pmod2$. If D is odd, every argument of $\mathcal{N}$ in (13) is odd, so every term is nonnegative by Lemma 3.2, and the term $j=0$ is $\mathcal{N}(D+2)>0$. If $D=2u$ is even, $-\mathcal{C}(2u)$ is the coefficient

of x^{u+1} in $f\prod_r(1-x^r)^{-e_r}$ with $f=\sum_{t\ge0}\big(-\mathcal{N}(2t)\big)x^t$. By Lemma 3.2, $f/2=-1+\sum_{t\ge1}a_tx^t$ with $a_t\ge48$, and Lemma 2.3 with $K=16$ gives $-\mathcal{C}(2u)>0$. The orbits follow from the $\Gamma_1(2)$ invariance of d^*. □

4. The second row: $Q^2=2$

4.1. **The closed form.** Let

$$f_6=\tfrac{5}{189}E_6(\tfrac{\sigma}{2})-\tfrac{68}{189}E_6(\sigma),\qquad f_4=\tfrac{4}{45}E_4(\tfrac{\sigma}{2})+\tfrac{26}{45}E_4(\sigma),\qquad f_2=-\tfrac{1}{3}\mathfrak{E}(\tfrac{\sigma}{2}),\tag{17}$$

and $J_2=f_6A^2+f_4AB+f_2B^2$, a weak Jacobi form of weight 2 and index 2 for $\Gamma^0(2)$. Let $\mathcal{H}|V_2$ be the image of $\mathcal{H}$ under the Hecke-type operator V_2 of [8, (4.37)],

$$\mathcal{H}|V_2=\sum_{n\ge0}\sum_{\ell\in\mathbb{Z}}C_2(8n-\ell^2)\,p^ny^\ell,\qquad C_2(\Delta)=H(\Delta)+2H(\Delta/4).\tag{18}$$

Proposition 4.1.

$$L_3+L_1L_2+\tfrac{1}{6}L_1^3=\tfrac{4}{27}\big(B^3-3E_4(\sigma)A^2B+2E_6(\sigma)A^3\big)+A\,J_2.\tag{19}$$

Proof. By Lemma 2.1 both sides are Jacobi forms of weight 0 and index three for $\Gamma(12)$, even and holomorphic in v and weak at every cusp; the functions $E_k(\sigma/2)$ and $\mathfrak{E}(\sigma/2)$ are modular for $\Gamma^0(2)\supset\Gamma(12)$. Let G be the difference and χ_ν its Taylor coefficients in v, as in the proof of Proposition 3.1. Now $[PSL_2(\mathbb{Z}):\bar\Gamma(12)]=576$ and the width of ∞ is 12, so by Sturm's theorem a modular form of weight ν for $\Gamma(12)$ vanishes once its coefficients vanish through $x^{8\nu}$. The coefficients of G vanish through x^{49} (we computed them in exact arithmetic for every monomial x^ey^ℓ with $e\le49$ and $|\ell|\le18$, which contains every monomial of these orders that a weak form of index three can carry), so $\chi_0=\chi_2=\chi_4=\chi_6=0$. Then $G(\sigma,\cdot)$ vanishes to order at least eight at $v=0$, while a nonzero function with the elliptic property of index three has six zeros in a period parallelogram [11, Theorem 1.2]. Hence $G=0$. □

Proposition 4.2. *In the $\mathbb{Z}_2$ CHL model*

$$g(\tfrac{\sigma}{2})\,\varphi_2=256\,\mathcal{A}_{2,2}+J_2-256\,\mathcal{H}|V_2,\tag{20}$$

so that $g(\frac{\sigma}{2})\varphi_2^F=J_2-256\,\mathcal{H}|V_2$.

Proof. Dabholkar, Murthy and Zagier [8, (9.11)] define

$$\varphi_{2,2}^{\rm opt}=\frac{B^3-3E_4A^2B+2E_6A^3}{1728\,A},$$

a meromorphic Jacobi form of weight 2 and index 2, and prove that its finite part $\varphi_{2,2}^{\rm opt}-\mathcal{A}_{2,2}$ equals $-\mathcal{H}|V_2$ [8, (9.13) and Theorem 9.1, proved in §10]. Divide (19) by A and use (7): since $\frac{4}{27}\cdot1728=256$,

$$g(\tfrac{\sigma}{2})\,\varphi_2=256\,\varphi_{2,2}^{\rm opt}+J_2=256\,\mathcal{A}_{2,2}-256\,\mathcal{H}|V_2+J_2.$$

The term $256\mathcal{A}_{2,2}/g(\sigma/2)$ is the polar part (8), as $\hat f(2)=256$. □

Remark 4.3. The index-two weak forms of weight two for $SL_2(\mathbb{Z})$ are spanned by E_6A^2 and E_4AB, whose coefficients at the discriminants $-4,-1$ are $(1,-4)$ and $(1,8)$; so a weak form of this space with no polar terms vanishes. This is why the optimal choice of [8] is forced at index two, and why the whole of the level-two dependence of φ_2^F sits in the holomorphic form J_2.

4.2. **The factorisation of J_2.** Put $u=\theta_2(\sigma)^4$ and $w=\theta_4(\sigma)^4$, so that $\theta_3(\sigma)^4=u+w$ by Jacobi's identity. The classical identities

$$\begin{aligned} E_4(\sigma)&=u^2+uw+w^2, \quad E_4(\tfrac{\sigma}{2})=16u^2+16uw+w^2, \quad \mathfrak{E}(\tfrac{\sigma}{2})=2u+w,\\ E_6(\sigma)&=w^3+\tfrac{3}{2}uw^2-\tfrac{3}{2}u^2w-u^3, \quad E_6(\tfrac{\sigma}{2})=w^3-30uw^2-96u^2w-64u^3 \end{aligned} \tag{21}$$

hold between modular forms for $\Gamma(4)$ of weight at most six; by Sturm's theorem each reduces to the first coefficients, which agree [20].[6] From (17),

$$f_2=-\tfrac{1}{3}(2u+w), \qquad f_4=2u^2+2uw+\tfrac{2}{3}w^2, \qquad f_6=-\tfrac{4}{3}u^3-2u^2w-\tfrac{4}{3}uw^2-\tfrac{1}{3}w^3. \tag{22}$$

Lemma 4.4. *For $k=2,3,4$,*

$$Y_k:=\frac{\theta_k(\sigma,v)^2}{\theta_k(\sigma)^2}=\frac{B}{12}-e_kA, \qquad e_2=-\frac{u+2w}{12}, \quad e_3=\frac{w-u}{12}, \quad e_4=\frac{2u+w}{12}.$$

Proof. The quartic relations of Jacobi [19, §21.2],

$$\theta_1(\sigma,v)^2\theta_4^2=\theta_3(\sigma,v)^2\theta_2^2-\theta_2(\sigma,v)^2\theta_3^2, \qquad \theta_1(\sigma,v)^2\theta_3^2=\theta_4(\sigma,v)^2\theta_2^2-\theta_2(\sigma,v)^2\theta_4^2,$$

with $\theta_k=\theta_k(\sigma)$, and $\eta^3=\frac{1}{2}\theta_2\theta_3\theta_4$ give $Y_3-Y_2=\theta_1(\sigma,v)^2\theta_4^2/(\theta_2^2\theta_3^2)=-\frac{1}{4}\theta_4^4\,A$ and $Y_4-Y_2=-\frac{1}{4}\theta_3^4A$, that is, $Y_3-Y_2=(e_2-e_3)A$ and $Y_4-Y_2=(e_2-e_4)A$. Together with $Y_2+Y_3+Y_4=B/4$ and $e_2+e_3+e_4=0$ this gives the lemma. □

Proposition 4.5.

$$J_2=\frac{16\,Y_4}{w}\Big[u(4u+w)\,Y_2-(u+w)(4u+3w)\,Y_3\Big]. \tag{23}$$

Equivalently, with $\theta_k=\theta_k(\sigma)$,

$$J_2=\frac{16\,\theta_4(\sigma,v)^2}{\theta_4^6}\Big[\theta_2^2\big(\theta_3^4+3\theta_2^4\big)\theta_2(\sigma,v)^2-\theta_3^2\big(3\theta_3^4+\theta_2^4\big)\theta_3(\sigma,v)^2\Big].$$

Proof. By Lemma 4.4, $B=12(Y_4+e_4A)$. Substituting into $J_2=f_6A^2+f_4AB+f_2B^2$,

$$J_2=\big(f_6+12e_4f_4+144e_4^2f_2\big)A^2+\big(12f_4+288e_4f_2\big)AY_4+144f_2Y_4^2.$$

By (22) the first bracket vanishes: its coefficients at u^3,u^2w,uw^2,w^3 are $-\frac{4}{3}+4-\frac{8}{3}$, $-2+6-4$, $-\frac{4}{3}+\frac{10}{3}-2$ and $-\frac{1}{3}+\frac{2}{3}-\frac{1}{3}$. The second bracket is $24u^2+24uw+8w^2-8(2u+w)^2=-8u(u+w)$, and $144f_2=-48(2u+w)$. So

$$J_2=-8\,Y_4\big[u(u+w)A+6(2u+w)Y_4\big].$$

By Lemma 4.4, $A=4(Y_2-Y_3)/w$ and $Y_4=\big((u+w)Y_3-uY_2\big)/w$. Inserting these, the coefficient of Y_2 in the bracket is $\big(4u(u+w)-6u(2u+w)\big)/w=-2u(4u+w)/w$ and that of Y_3 is $\big(-4u(u+w)+6(2u+w)(u+w)\big)/w=2(u+w)(4u+3w)/w$, which gives (23). The second form follows from $\theta_k^2\theta_k(\sigma,v)^2=\theta_k^4Y_k$, $\theta_2^4=u$, $\theta_3^4=u+w$ and $\theta_4^4=w$. □

[6] We checked each identity to x^{100} in exact arithmetic, far beyond the Sturm bound x^{12} for weight six on $\Gamma(4)$.

4.3. **Theta components.** A holomorphic Jacobi form ϕ of index two is determined by its theta components: $\phi = \sum_{\mu \bmod 4} G_\mu(\sigma)\,\vartheta_{2,\mu}(\sigma,v)$ with $\vartheta_{2,\mu}(\sigma,v) = \sum_{\ell\equiv\mu\,(4)} p^{\ell^2/8} y^\ell$ [11, §5]. Since J_2 is even, $G_1 = G_3$. Put

$$a_4 = \theta_3(4\sigma) = \sum_n x^{4n^2}, \qquad b_4 = \theta_2(4\sigma) = \sum_n x^{(2n+1)^2},$$
$$a_2 = \theta_3(2\sigma), \qquad b_2 = \theta_2(2\sigma), \qquad \psi = \sum_{j\in\mathbb{Z}} x^{(4j+1)^2/4}.$$

Then $\theta_3(\sigma) = a_4 + b_4$, $\theta_4(\sigma) = a_4 - b_4$, $a_2^2 = a_4^2 + b_4^2$, $b_2^2 = 2a_4b_4$ and $\theta_2(\sigma)^2 = 2a_2b_2$.

Lemma 4.6. *The theta components of J_2 are*

$$G_0 = -\frac{16\,a_4\big(3a_4^6 + 28a_4^5b_4 + 3a_4^4b_4^2 + 112a_4^3b_4^3 + 25a_4^2b_4^4 + 84a_4b_4^5 + b_4^6\big)}{(a_4-b_4)^4},$$
$$G_2 = -\frac{16\,b_4\big(a_4^6 + 84a_4^5b_4 + 25a_4^4b_4^2 + 112a_4^3b_4^3 + 3a_4^2b_4^4 + 28a_4b_4^5 + 3b_4^6\big)}{(a_4-b_4)^4},$$
$$G_1 = G_3 = \frac{16\,\psi\,\theta_2(\sigma)^2\big(a_4^4 + 28a_4^3b_4 + 6a_4^2b_4^2 + 28a_4b_4^3 + b_4^4\big)}{(a_4-b_4)^4}.$$

In particular $-G_0$, $-G_2$ and G_1 have nonnegative coefficients as series in $x^{1/4}$.

Proof. Expanding the squares, with $\vartheta_{1,\mu} = \vartheta_{1,\mu}(\sigma,v) = \sum_{r\equiv\mu\,(2)} p^{r^2/4} y^r$,

$$\theta_3(\sigma,v)^2 = a_2\vartheta_{1,0} + b_2\vartheta_{1,1}, \qquad \theta_4(\sigma,v)^2 = a_2\vartheta_{1,0} - b_2\vartheta_{1,1}, \qquad \theta_2(\sigma,v)^2 = b_2\vartheta_{1,0} + a_2\vartheta_{1,1}:$$

for the first, write $\frac12(k^2+k'^2) = \frac14(r^2+s^2)$ with $r = k+k'$, $s = k-k' \equiv r \pmod 2$, and sum over s; the others are the same computation with the sign $(-1)^r$ or with $k,k' \in \mathbb{Z}+\frac12$. In the same way

$$\vartheta_{1,0}^2 = a_4\vartheta_{2,0} + b_4\vartheta_{2,2}, \qquad \vartheta_{1,1}^2 = b_4\vartheta_{2,0} + a_4\vartheta_{2,2}, \qquad \vartheta_{1,0}\vartheta_{1,1} = \psi\,(\vartheta_{2,1} + \vartheta_{2,3}).$$

Hence

$$\theta_4(\sigma,v)^2\theta_2(\sigma,v)^2 = a_2b_2(a_4-b_4)(\vartheta_{2,0} - \vartheta_{2,2}) + (a_4-b_4)^2\psi\,(\vartheta_{2,1} + \vartheta_{2,3}),$$
$$\theta_4(\sigma,v)^2\theta_3(\sigma,v)^2 = a_4(a_4-b_4)(a_4+b_4)\,\vartheta_{2,0} - b_4(a_4-b_4)(a_4+b_4)\,\vartheta_{2,2}.$$

Insert these in the second form of Proposition 4.5, with $\theta_2(\sigma)^2 = 2a_2b_2$, $\theta_2^4 = 8a_4b_4(a_4^2+b_4^2)$, $\theta_3^4 = (a_4+b_4)^4$ and $\theta_4^4 = (a_4-b_4)^4$. The coefficient of $\vartheta_{2,1}$ is $16\psi\theta_2(\sigma)^2(\theta_3^4+3\theta_2^4)/\theta_4^4$, which is the stated G_1. The coefficients of $\vartheta_{2,0}$ and $\vartheta_{2,2}$ are

$$\frac{16\big[2a_2^2b_2^2(\theta_3^4+3\theta_2^4) - a_4(a_4+b_4)^3(3\theta_3^4+\theta_2^4)\big]}{(a_4-b_4)^5},$$
$$\frac{16\big[b_4(a_4+b_4)^3(3\theta_3^4+\theta_2^4) - 2a_2^2b_2^2(\theta_3^4+3\theta_2^4)\big]}{(a_4-b_4)^5},$$

and expanding the numerators and cancelling one factor $a_4 - b_4$ gives G_0 and G_2. Finally a_4, b_4, a_2, b_2, ψ have nonnegative coefficients, and

$$\frac{1}{\theta_4(\sigma)} = \frac{1}{a_4-b_4} = \prod_{n\ge1}\frac{1+x^n}{1-x^n}$$

has positive coefficients, which gives the sign statement. □

Write $G_\mu = \sum_D g_\mu(D)\, x^{D/4}$, so that $J_2 = \sum_{e,\ell} g_{\ell \bmod 4}(4e-\ell^2)\, x^e y^\ell$. In the same way $\mathcal{H}|V_2$ has the coefficient $C_2(4e-\ell^2)$ at $x^e y^\ell$ when e is even and 0 when e is odd. Since $\ell^2 \equiv 0, 4, 1 \pmod 8$ for $\ell \equiv 0, 2, \pm 1 \pmod 4$, the parity of e is determined by $D = 4e - \ell^2$ and $\mu = \ell \bmod 4$. So the numerator $J_2 - 256\,\mathcal{H}|V_2$ of φ_2^F has the coefficient

$$N_\mu(D) = g_\mu(D) - 256\,\epsilon_\mu(D)\,C_2(D), \qquad \epsilon_\mu(D) = \begin{cases} 1, & (D+\mu^2)/4 \text{ even}, \\ 0, & \text{otherwise}, \end{cases} \tag{24}$$

at $x^e y^\ell$, with $D = 4e - \ell^2$ and $\mu \equiv \ell \pmod 4$, $N_3 = N_1$. By Proposition 4.2 and (1),

$$c_2^F(n,\ell) = \sum_{j \ge 0} \beta(j)\, N_{\ell \bmod 4}(D + 4 - 4j), \qquad D = 4n - \ell^2. \tag{25}$$

For $Q^2 = 2$ this D is $Q^2P^2 - (Q\cdot P)^2$, in agreement with (9).

Lemma 4.7. (i) $N_\mu(D) = 0$ *for* $D < 0$.
(ii) $N_1(D) > 0$ *for every* $D \equiv 3 \pmod 4$, $D \ge 3$.
(iii) $N_2(0) = 0$ *and* $N_2(D) < 0$ *for* $D \equiv 0 \pmod 4$, $D \ge 4$.
(iv) $N_0(0) = 16$ *and* $-N_0(4t) \ge 1088$ *for* $t \ge 1$.

Proof. We use $a_4 = 1 + O(x^4)$, $b_4 = 2x + O(x^9)$, $\psi = x^{1/4} + O(x^{9/4})$, $\theta_2(\sigma)^2 = 4x^{1/2} + O(x^{5/2})$, all with nonnegative coefficients, and $1/(a_4 - b_4)^4 \ge 1/(1-x)^4$ coefficientwise. Here $F \ge F'$ means that $F - F'$ has nonnegative coefficients.

(i) By Lemma 4.6, G_0, G_1, G_2 begin with x^0, $x^{3/4}$, x^1, so $g_\mu(D) = 0$ for $D < 0$; and $C_2(D) = 0$ for $D < 0$.

(ii) For $D \equiv 3 \pmod 4$, $D/4$ is not an integer, so $C_2(D) = H(D)$. Keeping the terms a_4^4 and $28a_4^3 b_4$ of the numerator of G_1,

$$G_1 \;\ge\; 64\,x^{3/4}\Big(\frac{1}{(1-x)^4} + \frac{56x}{(1-x)^4}\Big), \qquad g_1(4k+3) \;\ge\; 64\Big[\binom{k+3}{3} + 56\binom{k+2}{3}\Big].$$

At $D = 3$, $(D+1)/4 = 1$ is odd, so $\epsilon_1(3) = 0$ and $N_1(3) = g_1(3) = 64$. For $k \ge 1$, $g_1(4k+3) \ge 3584k$, while $256\,H(4k+3) \le 256(4k+3) < 3584k$; so $N_1(4k+3) > 0$.

(iii) At $D = 0$, $g_2(0) = 0$ and $\epsilon_2(0) = 0$ because $(0+4)/4 = 1$ is odd. For $D = 4t$, $t \ge 1$, keeping the term a_4^6 of the numerator of G_2 gives $-G_2 \ge 32x/(1-x)^4$, so $-g_2(4t) \ge 32\binom{t+2}{3} > 0$, and $-256\,\epsilon_2 C_2(4t) \le 0$ because $C_2(D) \ge 0$ for $D > 0$.

(iv) At $D = 0$, $g_0(0) = -48$ and $\epsilon_0(0) = 1$, $C_2(0) = 3H(0) = -\frac14$, so $N_0(0) = -48 + 64 = 16$. For $t \ge 1$, keeping $3a_4^6$ and $28a_4^5 b_4$ in the numerator of G_0,

$$-G_0 \;\ge\; \frac{48 + 896x}{(1-x)^4}, \qquad -g_0(4t) \;\ge\; 48\binom{t+3}{3} + 896\binom{t+2}{3} \;\ge\; 1088,$$

and $-N_0(4t) = -g_0(4t) + 256\,\epsilon_0 C_2(4t) \ge -g_0(4t)$. □

The first values are $N_0(0), N_0(4), N_0(8) = 16, -1280, -9536$; $N_2(4), N_2(8) = -160, -5632$; $N_1(3), N_1(7), N_1(11) = 64, 3840, 32960$. Figure 1 shows the pattern of Lemma 4.7.

Theorem 4.8. *In the* $\mathbb{Z}_2$ *CHL model,* $d^*(Q,P) > 0$ *for every charge with* $Q^2 = 2$, $P^2 > 0$ *and* $2P^2 > (Q\cdot P)^2$, *and for every charge in the* $\Gamma_1(2)$ *orbit of such a charge.*

Proof. By Lemma 2.2 and (11), $d^* = (-1)^{\ell+1} c_2^F(n,\ell)$, and c_2^F is given by (25) with $D = 4n - \ell^2 > 0$; every $\beta(j)$ is positive.

If ℓ is odd, then $D \equiv 3 \pmod 4$, every argument $D + 4 - 4j$ is $\equiv 3 \pmod 4$, every term is nonnegative by Lemma 4.7(i), (ii), and the term $j = 0$ is $N_1(D+4) > 0$. So $d^* = c_2^F > 0$.

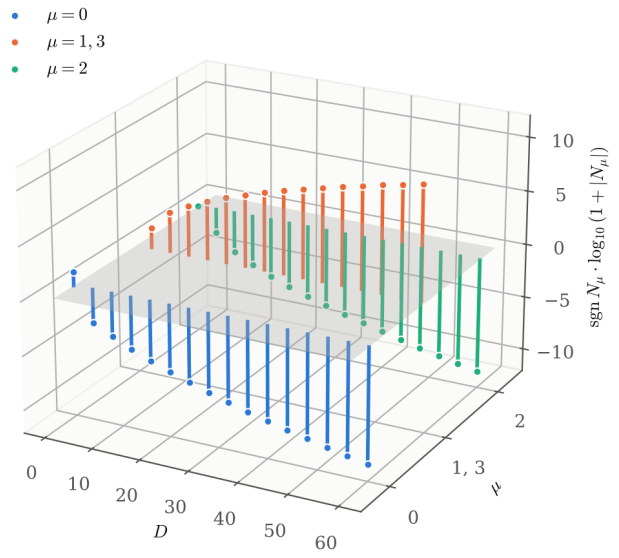


FIGURE 1. The coefficients $N_\mu(D)$ of the numerator of φ_2^F for $0 \le D \le 60$, plotted as $\operatorname{sgn} N_\mu \cdot \log_{10}(1+|N_\mu|)$ over the plane $N = 0$. The component $\mu = 0$ is positive only at $D = 0$, the component $\mu = 2$ is never positive, and the odd components are always positive.

If $\ell \equiv 2 \pmod 4$, then $D \equiv 0 \pmod 4$, every term is ≤ 0 by Lemma 4.7(i), (iii), and the term $j = 0$ is $N_2(D+4) < 0$ since $D + 4 \ge 8$. So $d^* = -c_2^F > 0$.

If $\ell \equiv 0 \pmod 4$, write $D = 4u$ with $u \ge 1$. Then $-c_2^F$ is the coefficient of x^{u+1} in $f \prod_r (1-x^r)^{-e_r}$, with $f = \sum_{t \ge 0} \big(-N_0(4t) \big) x^t$. By Lemma 4.7(iv), $f/16 = -1 + \sum_{t \ge 1} a_t x^t$ with $a_t \ge 68 > 16$, and Lemma 2.3 with $K = 16$ gives $-c_2^F > 0$, so $d^* > 0$. The orbits follow from the $\Gamma_1(2)$ invariance of d^*. □

5. VALUES AND COMPARISON WITH THE PRODUCT FORMULA

The first single-centred indices given by (13) and (25) are listed in Table 1. We expanded the product (3) in exact rational arithmetic and compared it with (12) and (20) on every coefficient $x^e y^\ell$ with $e \le 49$ and $|\ell| \le 18$; the two sides agree. Every entry of Table 1 that lies in Sen's Table 2 [16] coincides with it: at $Q^2 = 1$ the entries 360, 2164, 4352, 18944, 36024, 125860 and the value 2164 at $(P^2, Q \cdot P) = (6, 2)$, and at $Q^2 = 2$ the entries 198144, 67008, 6912 of the row $P^2 = 4$ and 1580672, 671744, 101376, 4352 of the row $P^2 = 6$.[7] Figure 2 shows $\log_{10} d^*$ on the two rows.

Remark 5.1. At $Q^2 = 3$, where $\hat{f}(3) = 1122$, the coefficients of the product formula that we computed (all $x^e y^\ell$ with $e < 60$) are matched by $g(\frac{\sigma}{2})\varphi_3 = 1122\,\varphi_{2,3}^{\mathrm{opt}} + J_3 = 1122\,\mathcal{A}_{2,3} - 1122\,\mathcal{H}|V_3 + J_3$, with $\varphi_{2,3}^{\mathrm{opt}}$ the form of [8, (9.12)–(9.13)] and J_3 a unique holomorphic Jacobi form of weight two and index three built from A, B and Eisenstein series at σ and $\sigma/2$. We have neither proved this identity, which needs the analogue of Proposition 4.1 at index four,

[7] In these entries the charge lies in the cone (3.9) of [16], and the index of the full partition function equals the single-centred index. The entries of Sen's table with $D \le 0$, such as -2 at $(Q^2, P^2, Q \cdot P) = (1, 2, 2)$ or -16 at $(2, 6, 4)$, are outside the scope of the theorem, as they should be.

$Q^2=1$		$Q^2=2$					
D	d^*	D	d^* (ℓ odd)	D	d^* ($\ell\equiv 0$)	D	d^* ($\ell\equiv 2$)
1	360	3	4352	4	18944	4	6912
2	2164	7	67008	8	198144	8	101376
3	4352	11	671744	12	1580672	12	992256
4	18944	15	5189696	16	10586880	16	7550400
5	36024	19	33581312	20	62035968	20	48253952
6	125860	23	190095232	24	327011328	24	270195712
7	233080	27	968248576	28	1578794128	28	1362534400
8	705536	31	4522209472	32	7075522048	32	6304638848

TABLE 1. The first single-centred indices on the rows $Q^2=1$ and $Q^2=2$, as functions of $D=Q^2P^2-(Q\cdot P)^2$ and, at $Q^2=2$, of $\ell=Q\cdot P$ modulo 4.

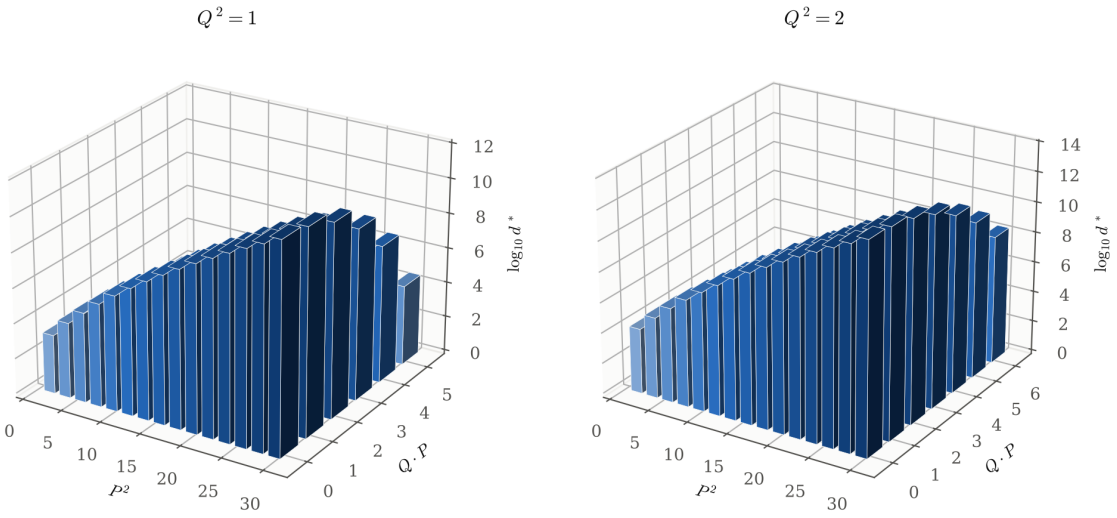


FIGURE 2. $\log_{10} d^*(Q,P)$ for $Q^2=1$ (left) and $Q^2=2$ (right), $2\le P^2\le 30$ and $0\le Q\cdot P\le \min(6,P^2)$ with $Q^2P^2>(Q\cdot P)^2$. Every bar is positive, as Theorems 3.3 and 4.8 assert.

nor determined the signs of the theta components of J_3; positivity at $Q^2=3$ therefore remains open, as does the conjecture for all larger Q^2. In every case the mock part of $g(\frac{\sigma}{2})\varphi_m^F$ is fixed by the shadow [4, 8], and what is needed is the sign pattern of the holomorphic remainder J_m.

DECLARATIONS

Data availability. The programs that expand the product formula (3) and check the identities of Propositions 3.1, 4.1 and 4.2, the identities (21) and the values of Table 1 in exact arithmetic are available from the author on request.

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Ethics statement. This study involves no human participants, animals or personal data; no ethical approval was required.

Use of AI tools. Claude Code (Opus 5.5 and Fable 5.1, Anthropic), GPT-6.1 Sol and GPT-6 Astra (OpenAI) were used to write and run the programs behind the computations and to help prepare the manuscript. The author checked every statement and proof and is responsible for the content.

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